$\therefore\ C(x) + 2x = \frac{x^3}{3} - 5x^2 + 28x + 10 + 2x$

P = Total revenue – (Total cost + tax)

$= (2530 - 5x)x - (\frac{x^3}{3} - 5x^2 + 28x + 10 + 2x)$

$= -\frac{x^3}{3} + 2500x - 10$

$\frac{dP}{dx} = -x^2 + 2500$ and $\frac{d^2P}{dx^2} = -2x$

$\frac{dP}{dx} = 0 \Rightarrow 2500 - x^2 = 0 \Rightarrow x^2 = 2500$

$\therefore\ x = 50$ (–50 is not acceptable)

At $x = 50\ \frac{d^2P}{dx^2} = -100 < 0$

P is maximum when $x = 50$.

$\therefore\ P =$ 2530 – 5(50) = ₹ 2280.

Example 6.29

The manufacturing cost of an item consists of ₹ 1,600 as over head material cost ₹ 30 per item and the labour cost ₹ $\left(\frac{x^2}{100}\right)$ for x items produced. Find how many items be produced to have the minimum average cost.

Solution:

As per given information for producing x units of certain item $C(x)$ = labour cost + material cost + overhead cost.

$= \frac{x^2}{100} + 30x + 1600$

$AC = \frac{C(x)}{x} = \frac{\frac{x^2}{100} + 30x + 1600}{x}$

$= \frac{x}{100} + 30 + \frac{1600}{x}$

$\frac{d(AC)}{dx} = \frac{1}{100} - \frac{1600}{x^2}$ & $\frac{d^2(AC)}{dx^2} = \frac{3200}{x^3}$

$\frac{d(AC)}{dx} = 0 \Rightarrow -\frac{1600}{x^2} + \frac{1}{100} = 0$

$\Rightarrow \frac{1}{100} = \frac{1600}{x^2} \Rightarrow x^2 = 160000$

$\therefore\ x = 400$ (–400 is not acceptable)

When $x = 400$, $\frac{d^2(AC)}{dx^2} = \frac{3200}{400^3} > 0$

AC is minimum at $x = 400$

Hence 400 items should be produced for minimum average cost.

Exercise 6.2

1. The average cost function associated with producing and marketing x units of an item is given by $AC = 2x - 11 + \frac{50}{x}$. Find the range of values of the output x, for which AC is increasing.
2. A television manufacturer finds that the total cost for the production and marketing of x number of television sets is $C(x) = 300x^2 + 4200x + 13,500$. If each product is sold for ₹ 8,400, show that the profit of the company is increasing.
3. A monopolist has a demand curve $x = 106 - 2p$ and average cost curve $AC = 5 + \frac{x}{50}$, where p is the price per unit output and x is the number of units of output. If the total revenue is $R = px$, determine the most profitable output and the maximum profit.
4. A tour operator charges ₹136 per passenger with a discount of 40 paisa for each passenger in excess of 100. The operator requires at least 100 passengers to operate the tour. Determine the number of passenger that will maximize the amount of money the tour operator receives.
5. Find the local minimum and local maximum of $y = 2x^3 - 3x^2 - 36x + 10$.
6. The total revenue function for a commodity is $R = 15x + \frac{x^2}{3} - \frac{1}{36}x^4$. Show that at the highest point average revenue is equal to the marginal revenue.

06_11th_BM-STAT_Ch-6-EM.indd 145 21-04-2020 12:22:08 PM